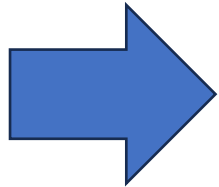


GSLAC

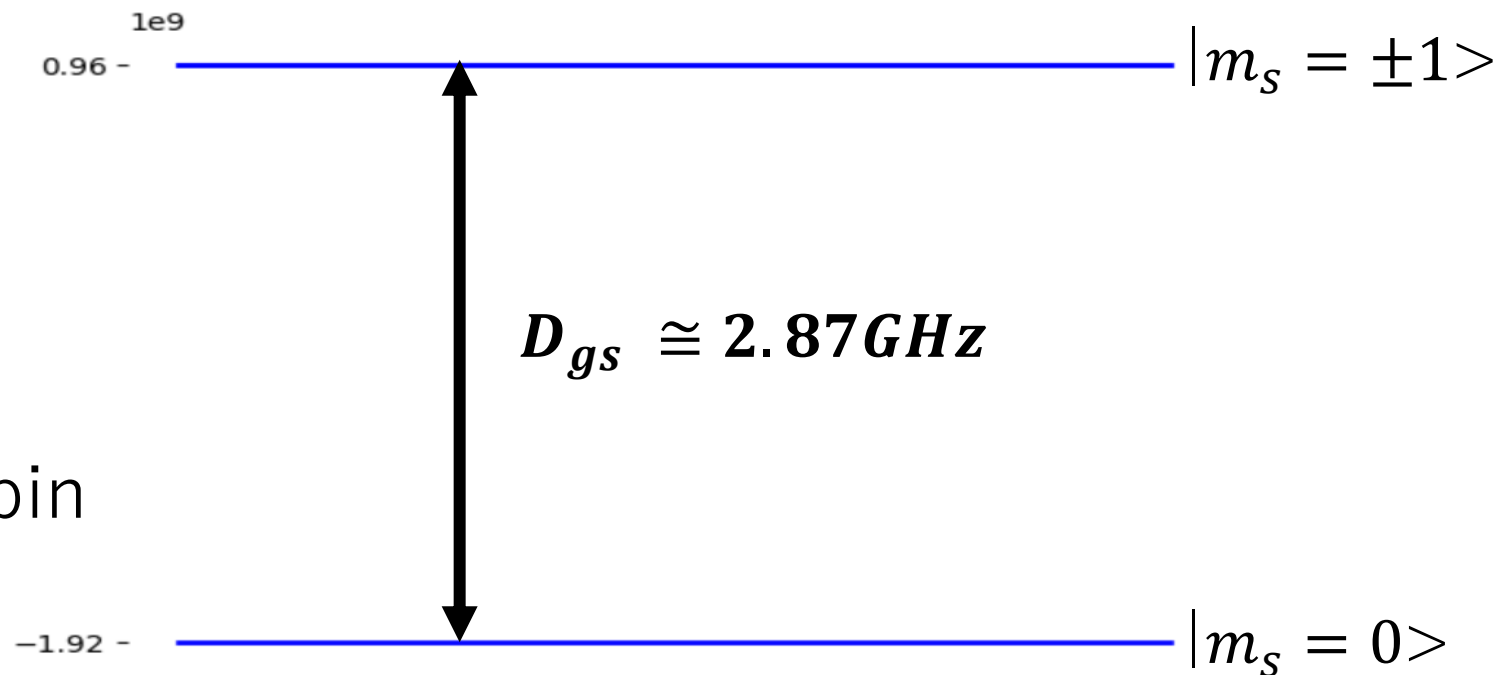
$$\hat{H}_{gs} = D_{gs} \left[\hat{S}_z^2 - \frac{S(S+1)}{3} \right]$$

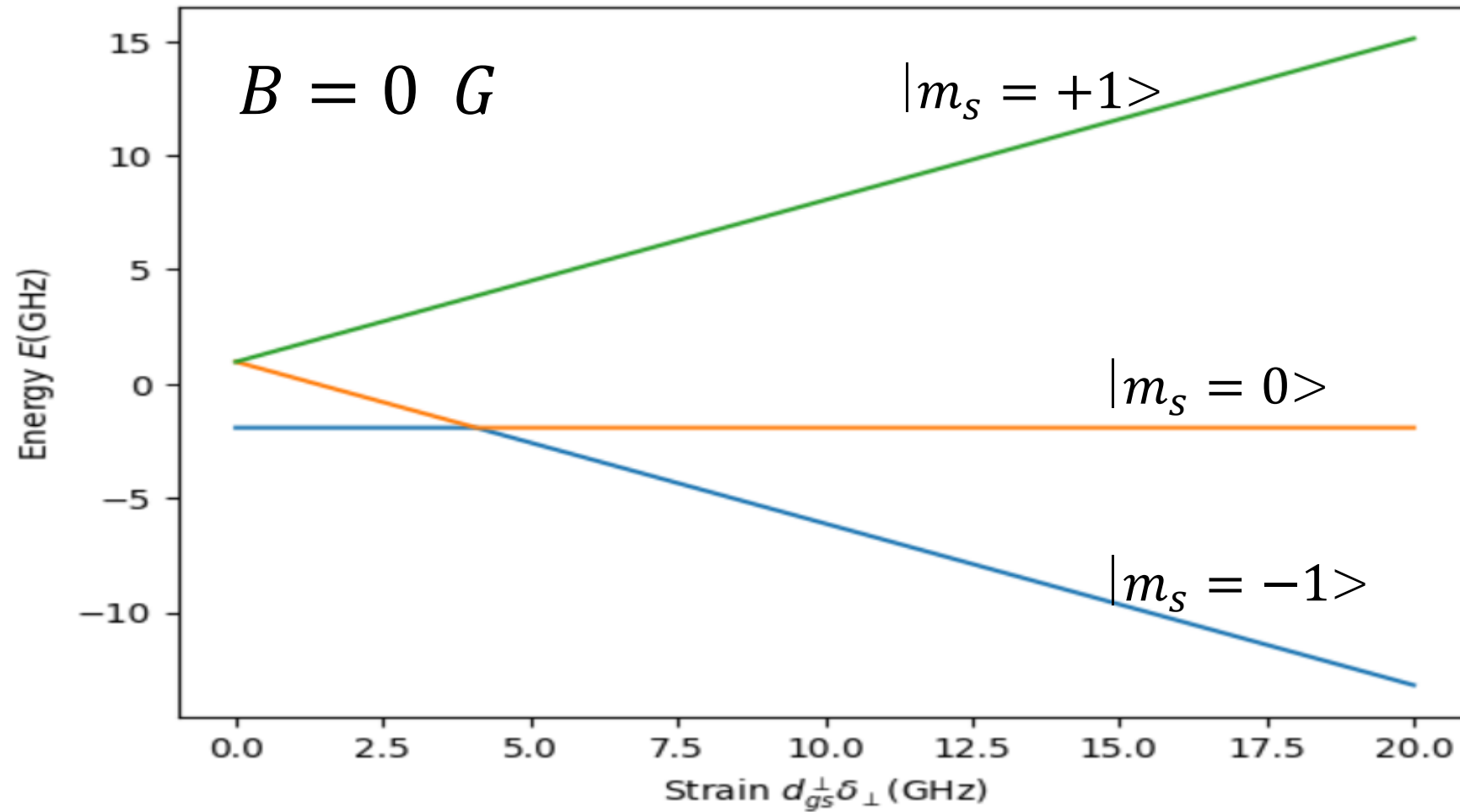
$$\begin{aligned} \hat{V}_{gs} = & \mu_B g_{gs}^{\parallel} \hat{S}_z B_z + \mu_B g_{gs}^{\perp} (\hat{S}_x B_x + \hat{S}_y B_y) + d_{gs}^{\parallel} (E_z + \delta_z) \left[\hat{S}_z^2 - \frac{S(S+1)}{3} \right] \\ & + d_{gs}^{\perp} (E_x + \delta_x) (\hat{S}_y^2 - \hat{S}_x^2) + d_{gs}^{\perp} (E_y + \delta_y) (\hat{S}_x \hat{S}_y - \hat{S}_y \hat{S}_x) \end{aligned}$$

(From Doherty 2013)

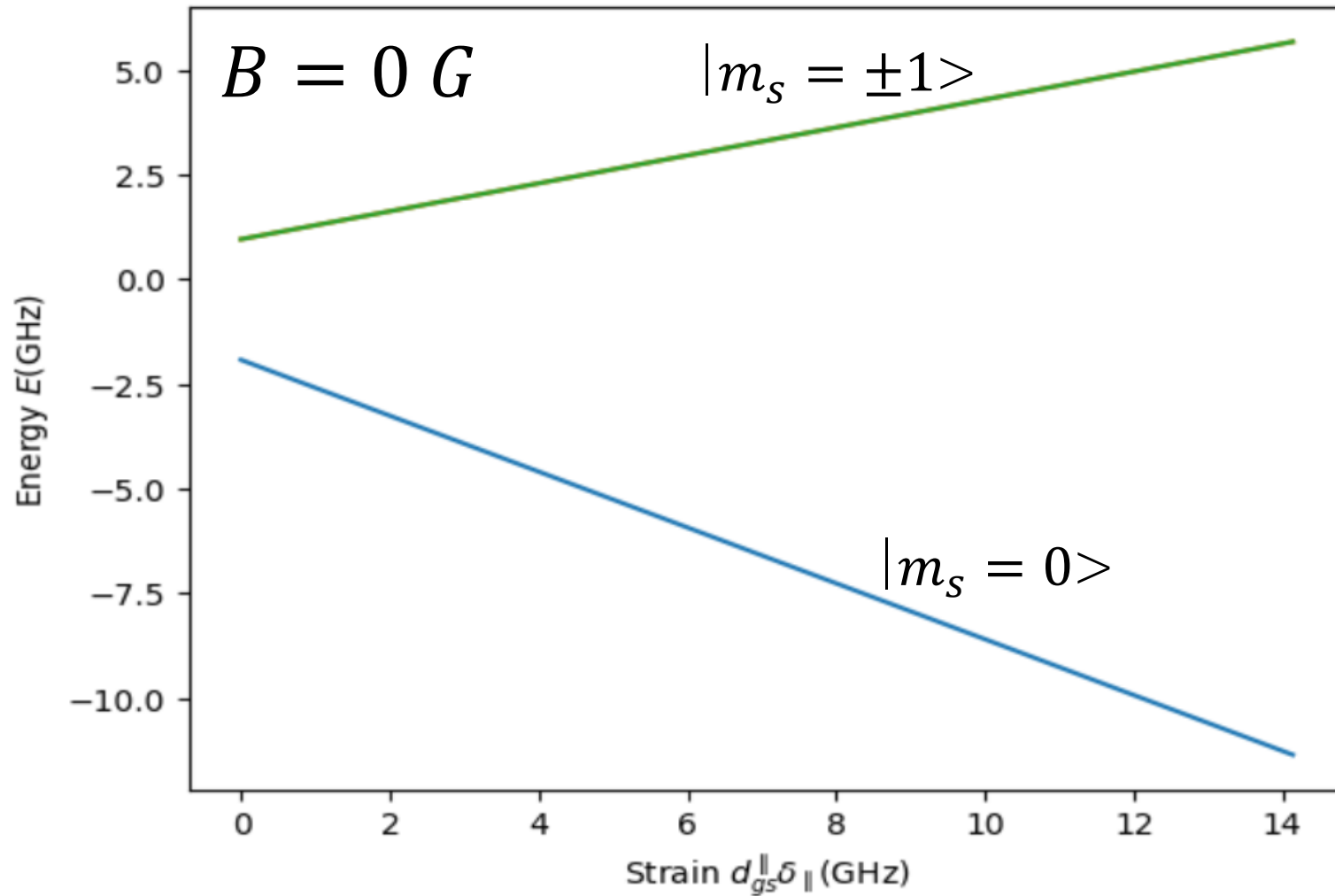


Temporarily, we ignore
the interaction by nuclear spin
for comparison to ESLAC

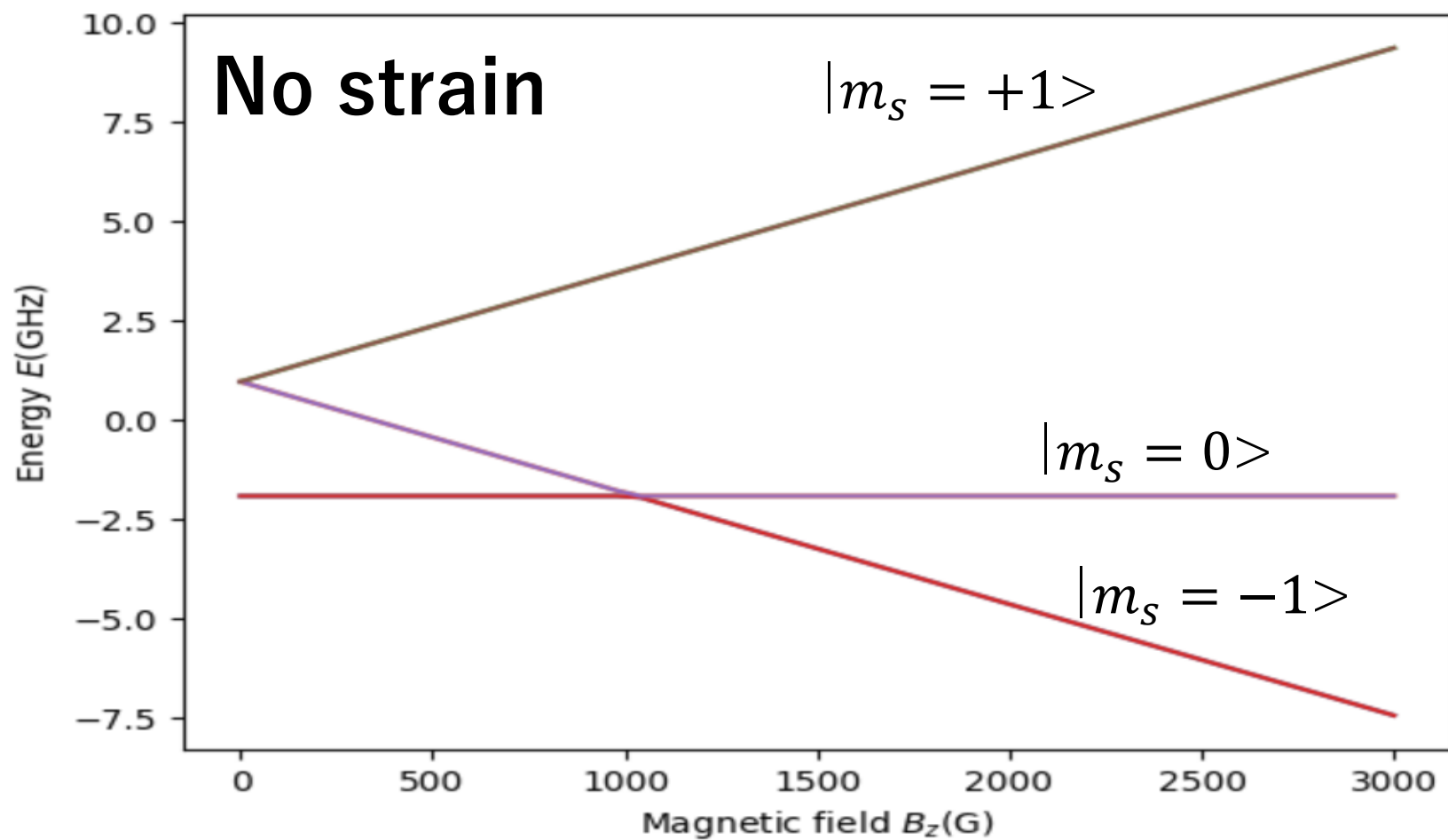


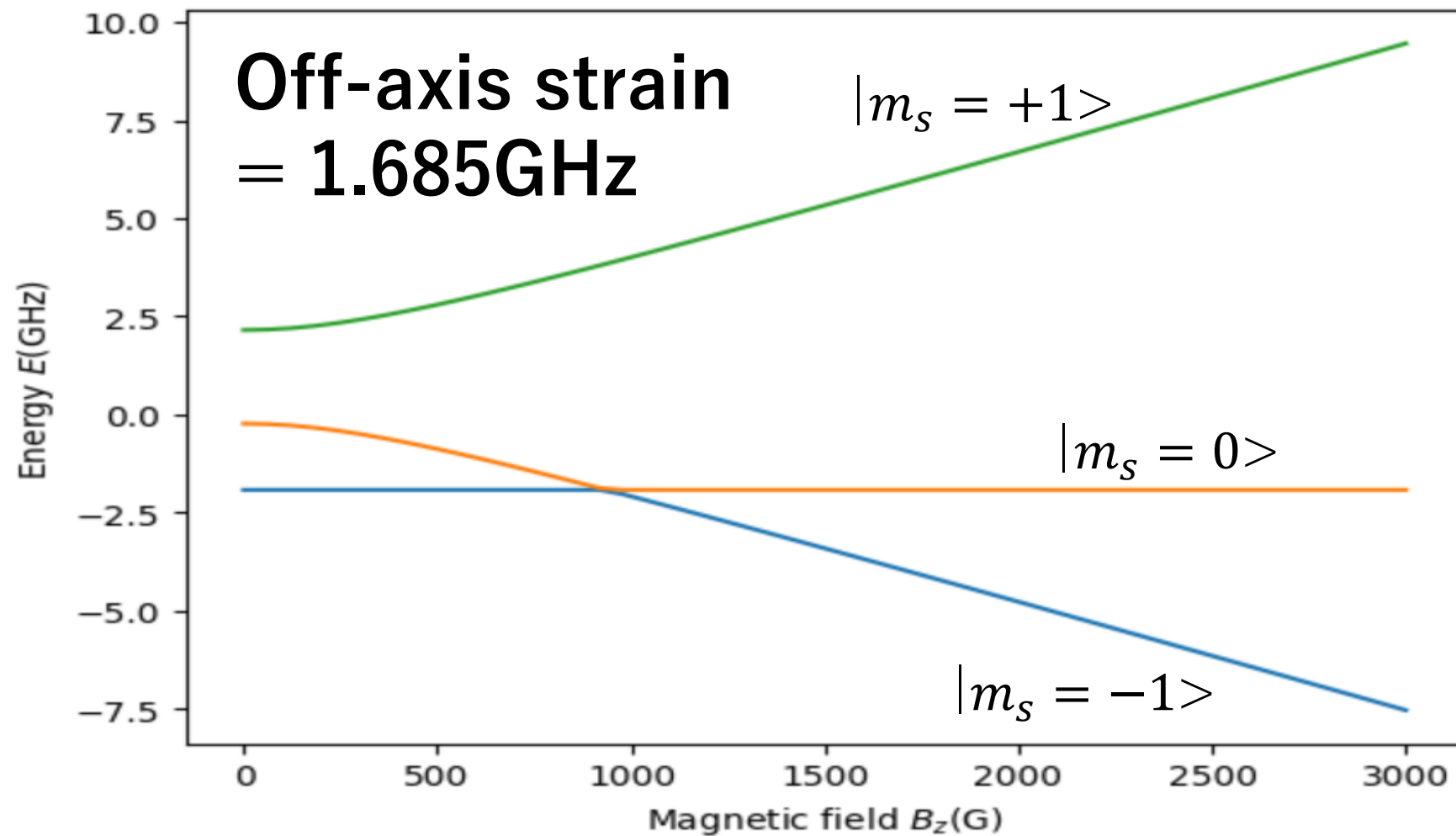


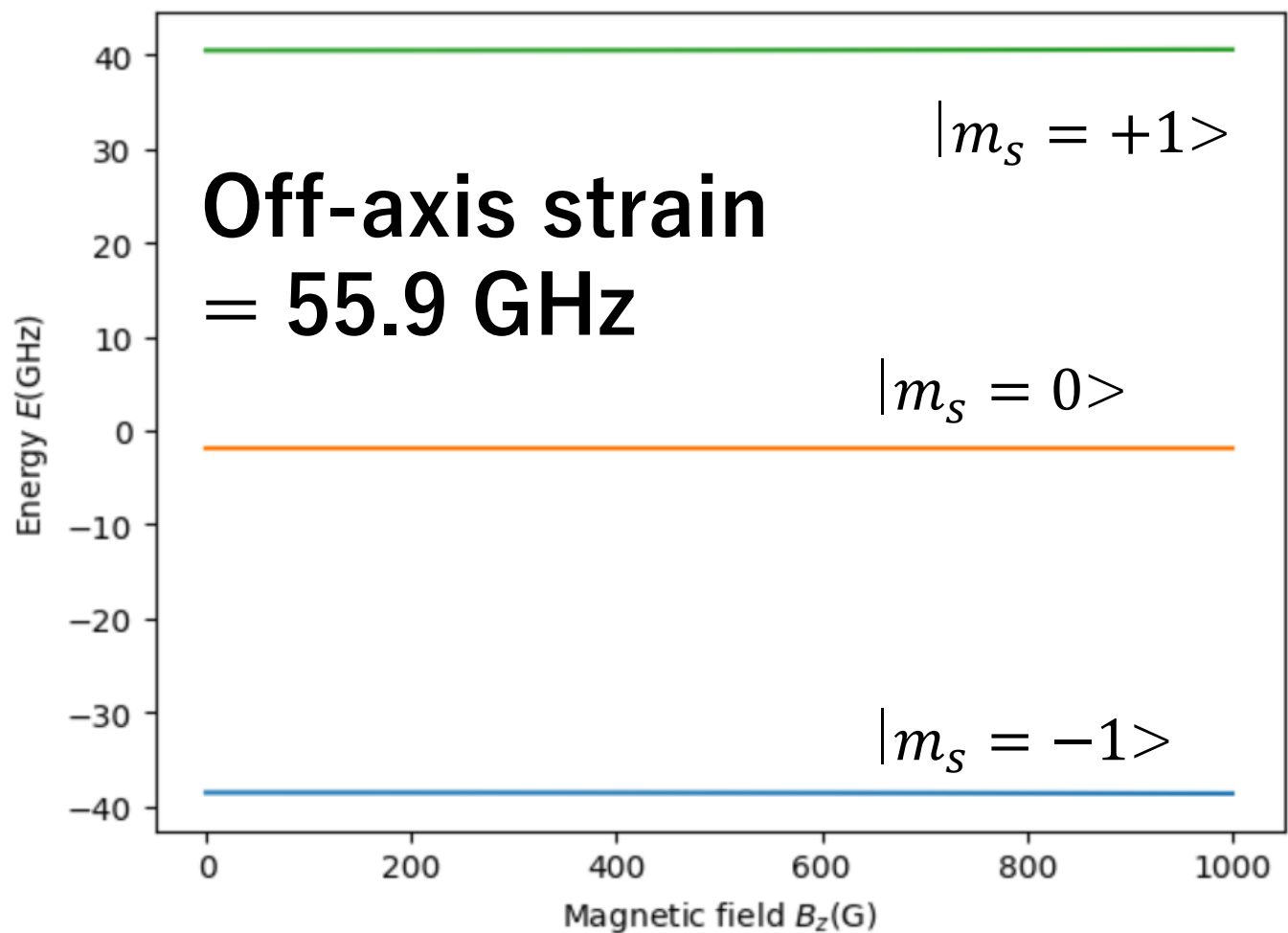
- Off-axis strain quantifies the energy splitting between $|m_s = +1\rangle, |m_s = -1\rangle$



- On-axis strain quantifies the energy splitting between $|m_s = \pm 1\rangle, |m_s = 0\rangle$







Summary

- Off-axis strain quantifies the energy splitting between $|m_s = +1\rangle, |m_s = -1\rangle$
- On-axis strain quantifies the energy splitting between $|m_s = \pm 1\rangle, |m_s = 0\rangle$

GSLAC points

Off axis strain $\delta^\perp$	On axis magnetic field B_z
~4 GHz	0 G
0 GHz	~1000G
1.685 GHz	~950G